

Technical Document #684: Database Systems - Comprehensive Overview

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Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Relational databases organize data into tables with rows and columns.
- Sharding distributes data across multiple database servers.
- Database caching stores frequently accessed data in memory.